

4(a)	any four from: 1 add DNA to give a new (named), characteristic / protein ; 2 <u>restriction</u>, enzyme / endonuclease, to obtain, gene / allele / section of DNA or <u>restriction</u>, enzyme / endonuclease, to cut plasmid ; 3 combine gene with / insert gene into, vector / plasmid ; 4 introduce, (recombinant) vector / plasmid, to, bacterium / (named host) cell ; 5 clone / multiply, (named, recombinant / GM) organism ; 6 AVP ;	4
4(b)(i)	1 stem cells, proliferate / divide / multiply / make new cells ; 2 (correct) gene is, inherited by / passed on to / present in, new cells ;	2
4(b)(ii)	any two from: 1 to switch gene on / so gene is expressed / allow transcription ; 2 at right time / all the time / in sufficient quantities ; 3 to allow binding of, RNA polymerase / transcription factors ;	2

Question	Answer	Marks
4(b)(iii)	any two from: 1 retrovirus / new gene, could / must not, insert in wrong place / disrupt other genes ; 2 could / must not, cause cancer ; 3 could / must not, cause, infection / disease ; 4 could / must not, cause, immune / allergic, response ; 5 AVP ;	2
4(c)	any three from: 1 HD is caused by a <u>dominant</u> allele ; 2 adding a 'normal' recessive allele would not work ; 3 gene editing can / we need to, delete / remove / inactivate, HD / mutant, allele / DNA ; 4 ref. (surplus CAG) repeats / triplets / stutter ;	3